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PRACTICAL CYBER DEFENSES FOR GRID-SCALE BATTERIES

An Overview of Threats and Protections

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Sandia National Laboratories

Native Computer Science & Cybersecurity Conference

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ABOUT ME



DR. VICTORIA A. O'BRIEN

2016

Texas Tech University



First Project - Rover



Student Activities Board



WIE & Faculty Dinner



Women in Engineering Officers



WIE & IEEE Arbor Day



IEEE Gameday Event

2020

B.S. in Electrical Engineering (2020)



DR. VICTORIA A. O'BRIEN



2021

M.S. in Electrical Engineering (2021)

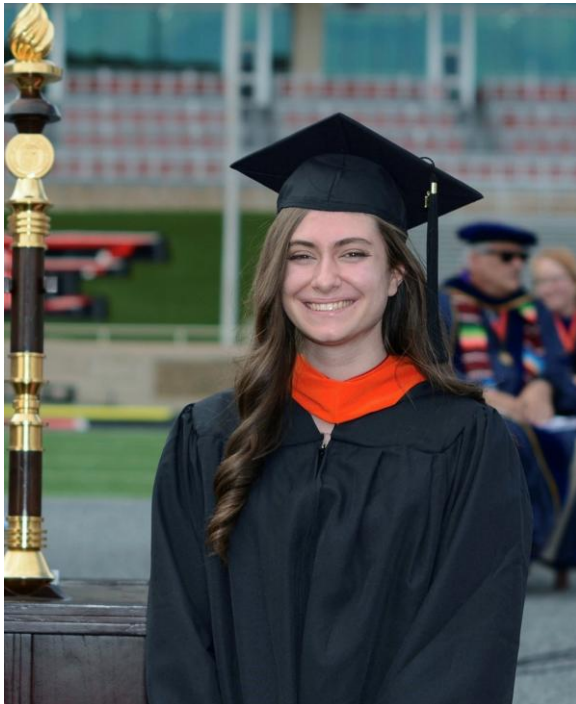
Intern at Sandia (2021- 2023)

2023

Ph.D. in Electrical Engineering (2023)

Full-time at Sandia (2023 - Present)

2026



Masters Graduation
May 2021



Ph.D. Defense
October 2023



Ph.D. Hooding
December 2023



Ph.D. Graduation
December 2023

BATTERY ENERGY STORAGE SYSTEM (BESS) OVERVIEW



COMPONENTS IN A BESS

- BESSs are complex cyber-physical systems with integrated systems and subsystems
- Combine components and software from multiple vendors across the globe

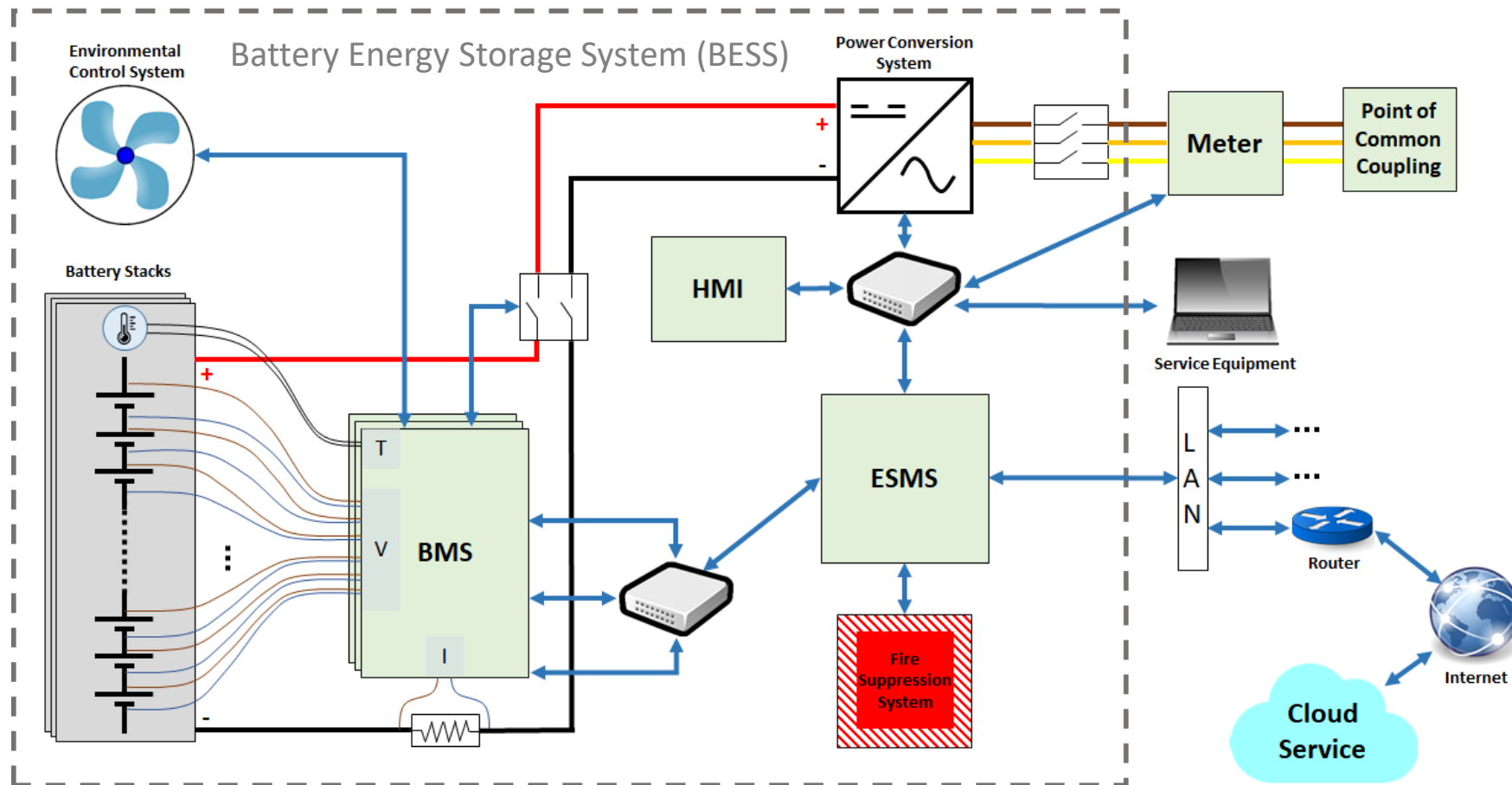
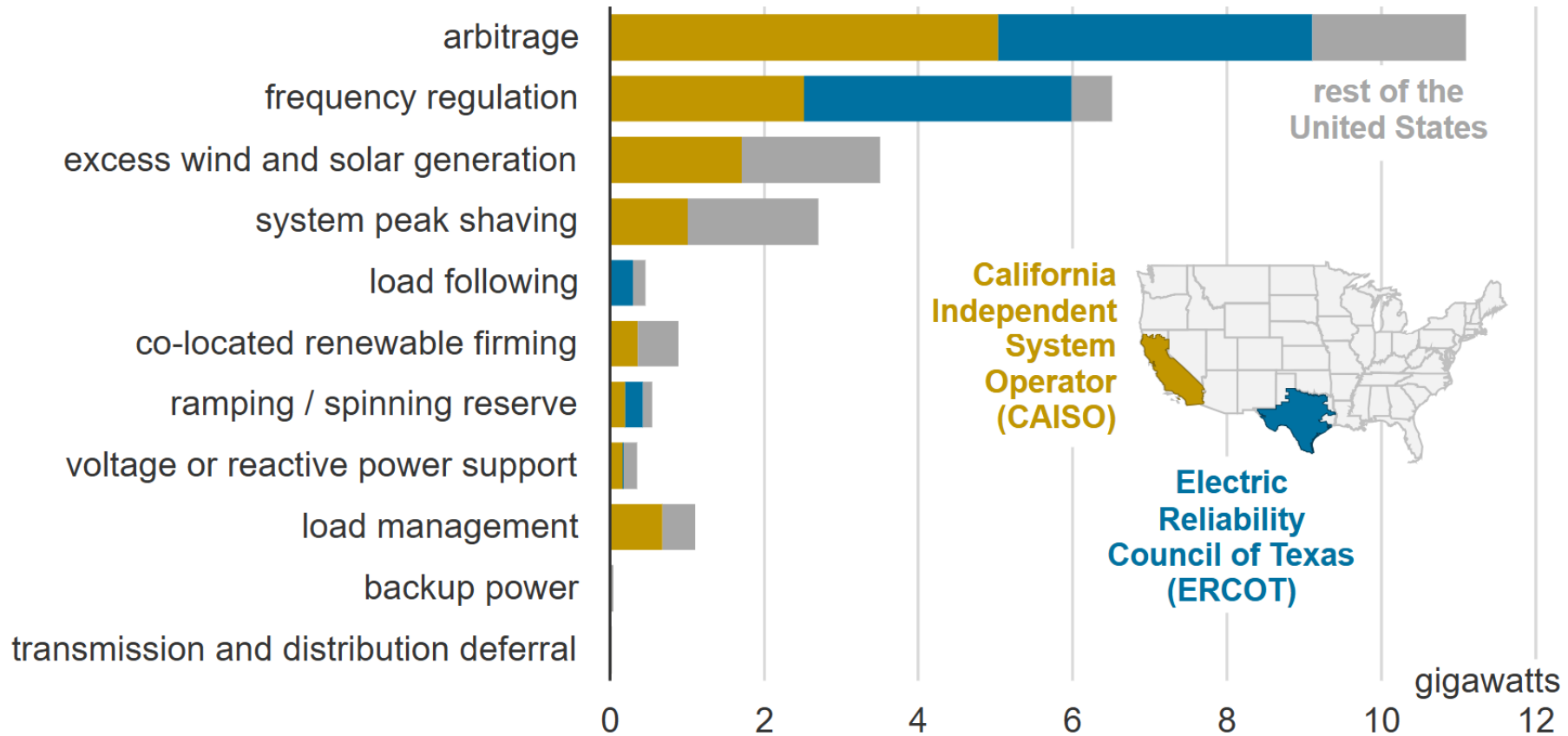


Diagram of a BESS from [1]

USES OF UTILITY-SCALE BESS



Primary use case of U.S. utility-scale battery capacity (2024)



Primary uses of BESS [2]

*I believe going forward we will see BESSs used to support data centers and load growth

MOSS LANDING BATTERY FIRES



- The September 2022 incident:
 - Water ingress caused overheating which led to fire [3]
 - Battery system was disconnected from the grid and remained offline for three months [3]
- The January 2025 incident:
 - Cause unknown [4]
 - Burning and destruction of many battery cells and release of smoke plume, fire lasted days



2025 Moss Landing fire
Photo from [4]

Foul play is not suspected in either of the Moss Landing incidents
Imagine the increased damage if these events had been caused by a bad actor

BESS THREAT EXEMPLARS

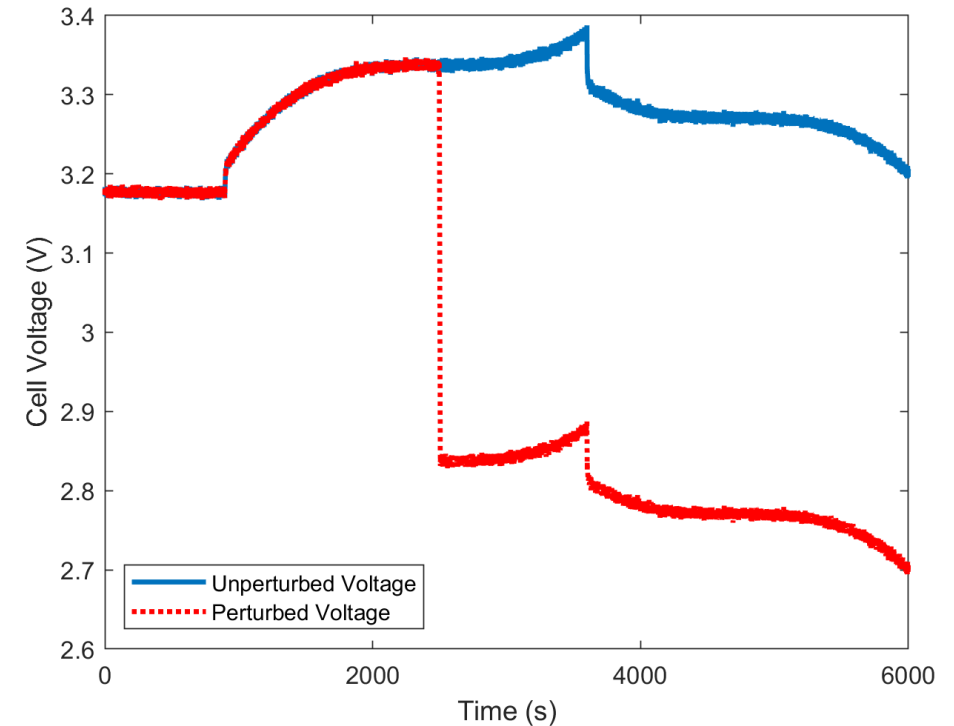


HYPOTHETICAL SCENARIOS

SCENARIO 1: FALSE DATA INJECTION ATTACK



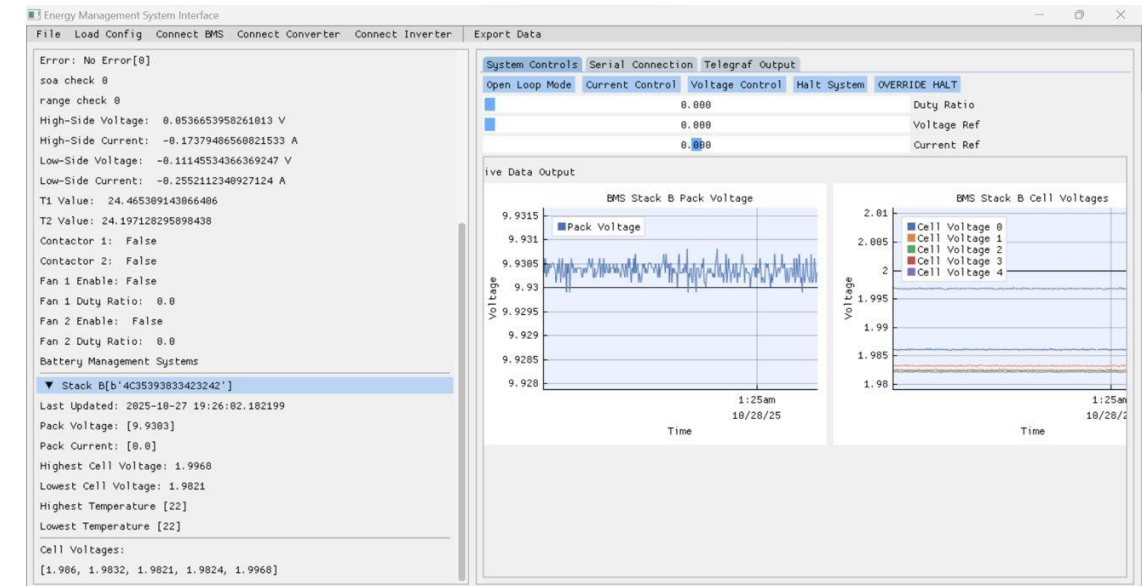
- Exemplar: False data injection attack (FDIA)
 - An adversarial nation hacks into the battery management system (BMS) of a BESS and corrupts the incoming voltage measurements
- Consequences: Corrupted control system
 - Battery degradation or damage
 - Overcharge/overdischarge
 - Overheating, thermal runaway, fire
- Similar incident: Stuxnet (2010)
 - Stuxnet worm sent false centrifuge readings to control system, causing the devices to spin out of control



Voltage following -500 mV FDIA

SCENARIO 2: MALICIOUS UPDATE

- Exemplar: Malicious software update
 - An energy management system company offers an update to their software via the cloud. Unfortunately, a bad actor has tampered with the code and removed the management systems' safety limits
- Consequences:
 - Corrupted safety subsystems
 - Overcharge/overdischarge
 - Overheating, thermal runaway, fire
- Similar incident: SolarWinds attack (2020)
 - Hackers inserted a malicious backdoor into SolarWinds' Orion network which was pushed to thousands of users, causing data theft and spying



Emulated energy management system dashboard containing safety checks
Photo courtesy of Armando Montoya

SCENARIO 3: RANSOMWARE / INSIDER THREAT



- Exemplar: BESS system held for ransom
 - While checking their email, a utility employee clicked on a malicious attachment. Ransomware was downloaded and the employee was locked out of their machine.
- Consequences:
 - Potential data leakage
 - Inability to access system
- Similar incident: Colonial Pipeline (2021)
 - DarkSide hackers obtained access to Colonial Pipeline's IT infrastructure, stole data, and locked access, demanding millions in payouts



Example of ransomware attack 'WannaCrypt'
Photo from [5]

SCENARIO 4: DENIAL OF SERVICE (DOS) ATTACK



- Exemplar: Denial of service (DoS)
 - Hackers flood the communication channels of a BESS, causing data loss and communication delays
- Consequences:
 - Missing readings
 - Time offline
 - Corrupted controls
- Similar incident: Solar farm DoS attack (2019)
 - A vulnerability in the firewall of a solar farm was exploited, causing the system to repeatedly reboot for 12 hours, communication outages, and reduced visibility

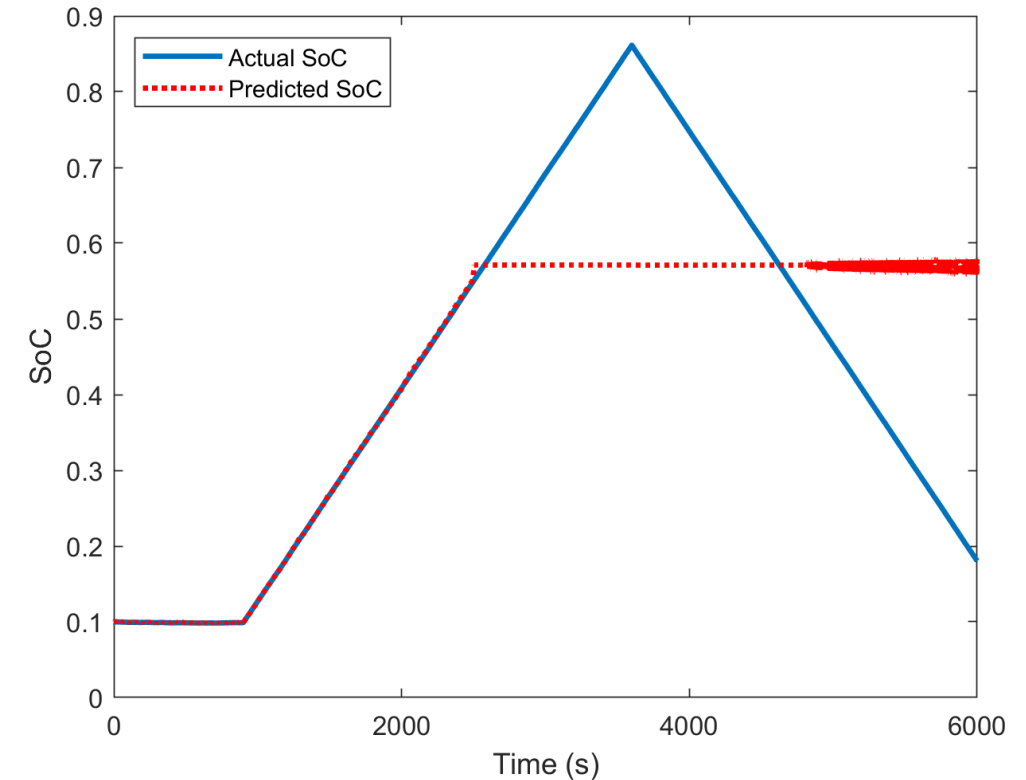


Missing temperature data due to
communications failure
Photos courtesy of Armando Montoya

SCENARIO 5: SUPPLY CHAIN



- Exemplar: Supply chain
 - A BESS is controlled by proprietary software. Bad actors hide a bug in the control software that corrupts Coulomb counting after one year of deployment
- Consequences: Corrupted control system
 - Battery degradation or damage
 - Overcharge/overdischarge
 - Overheating, thermal runaway, fire
- Similar incident: Pager explosions (2024)
 - Pagers were intercepted and tampered with by adding explosive devices near the Lithium-ion battery, explosives were then remotely detonated



Corrupted state of charge

SCENARIO 6: PHYSICAL DAMAGE

- Exemplar: BESS shooting
 - A bad actor uses firearms and explosive devices to destroy a BESS deployment
- Consequences:
 - BESS damage and destruction
 - Time offline
 - Degraded public trust
- Similar incident: Grid substation shootings (2013, 2020s)
 - Multiple substations shot up by bad actors across the U.S. throughout the 2020s. In addition, the infamous Metcalf Substation shooting of 2013



Footage from Metcalf shooting
Photo from [6]



THE BESS THREAT SURFACE



THREAT ACTOR GOALS



Advanced Persistent Threats

- Damaging systems
- Stealing intellectual property

Insider Threats

- Revenge / sabotage
- Leaking data
- Espionage
- Financial gain

Hacktivists

- Demonstrations
- Propaganda
- Fame
- Social change

Hackers

- Enhance skills
- Revenge
- Exploit vulnerabilities
- Financial gain
- Completing jobs

Vigilantes

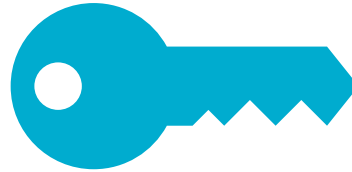
- Protect from unethical hackers

Threat actors and their goals [7]

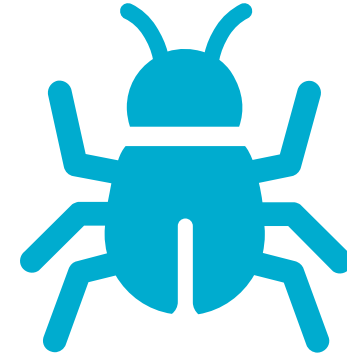
ENERGY SECTOR THREAT MODEL



Advanced Persistent Threat



Insider Threat



Hackers



Extreme Weather

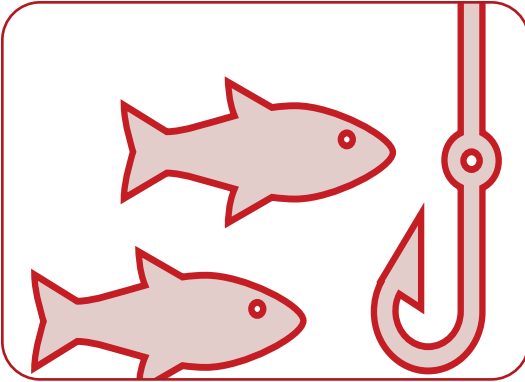


Physical Threat



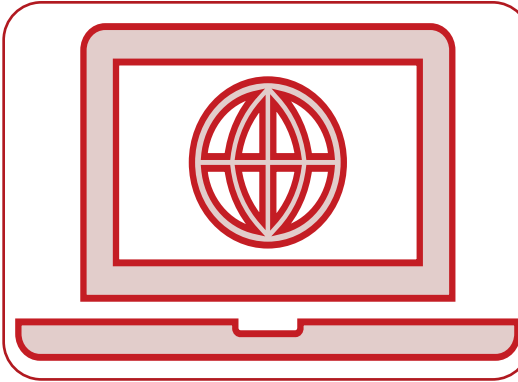
Hardware / Software Supply
Chain Threats

CONCERNS FROM ELECTRIC UTILITIES – CYBERATTACKS



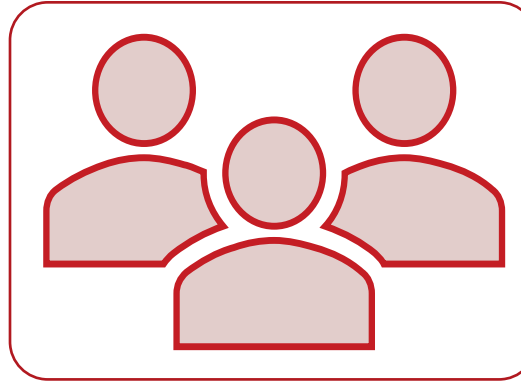
Phishing: using communications to obtain data or deploy malware

- compromised email
- spear phishing
- whaling
- vishing
- smishing



Malware: software intended to harm systems, obtain information, or obtain money

- viruses
- worms
- ransomware
- spyware



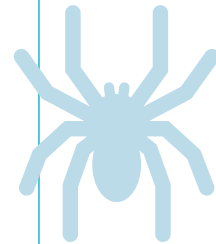
Social Engineering: manipulation technique to exploit human error

- social media phishing
- watering hole attack
- USB baiting
- physical social engineering

CONCERNS FROM ELECTRIC UTILITIES – ARTIFICIAL INTELLIGENCE



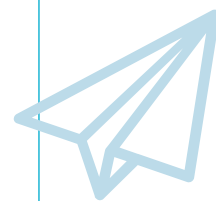
Using artificial intelligence to lower the barrier to entry for hackers



Using artificial intelligence to create more complex attacks



Using artificial intelligence to reverse engineer grid models and launch stealthy attacks



Using large language models to create more convincing phishing emails and social engineering attacks

CONCERNS FROM ELECTRIC UTILITIES – SUPPLY CHAIN



- 2014: Powerline Inc. counterfeit batteries sold to U.S. Government
- White labeling BESS components or control systems

PRESS RELEASE

**Former CEO Of Simi Valley Company
Guilty Of Bilking Navy By Selling \$2.6
Million In Knock-Off Batteries Used On
Aircraft Carriers And Subs**

Headline for sale of counterfeit batteries to
U.S. government
Photo from [9]

- Monopoly on BESS component supply chains, removal of CATL batteries



Ribbon cutting for CATL BESS deployment
Photo from [10]

DETECTING CYBERATTACKS IN BESS



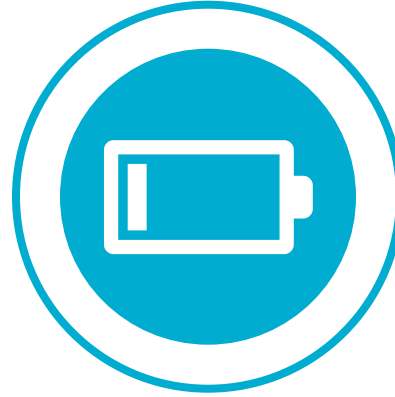
OUR WORK AT SANDIA

WHY DO WE NEED BATTERY MODELS?



Security

1. Intrusion detection
2. False data injection attacks
3. Attacks on control systems



Longevity

1. Modeling degradation
2. Estimation of state of health
3. Optimal controls



Safety

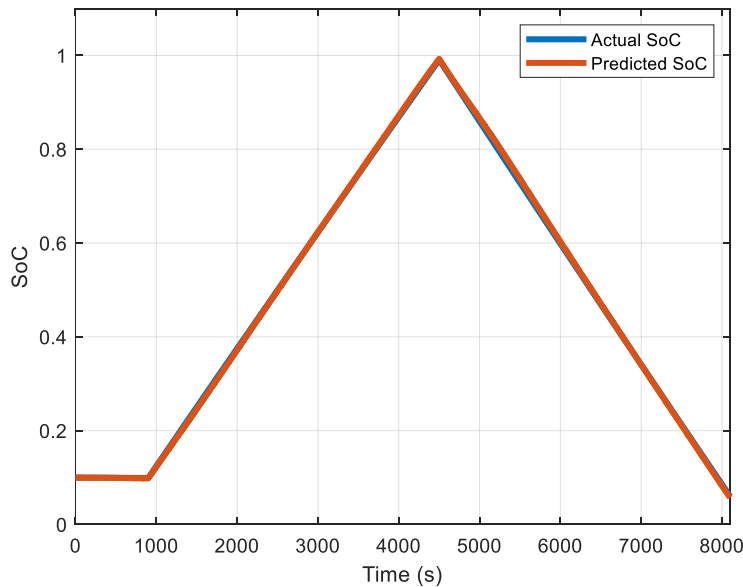
1. Fault / failure detection
2. Fire detection
3. Optimal controls

Research Goal: Use battery modeling, estimation, and statistics to flag naturally occurring (faults, failures) and manmade (cyberattacks) anomalies on battery sensors and to improve battery controls.

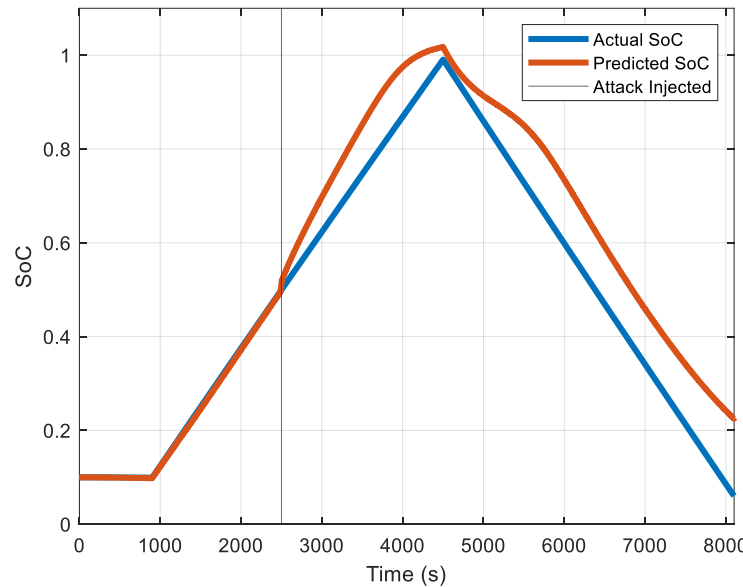
FALSE DATA INJECTION ATTACKS (FDIAs)



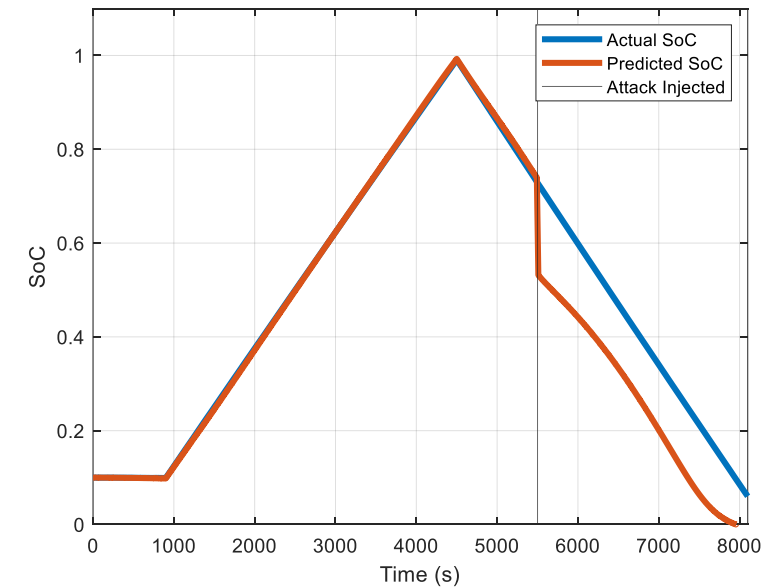
- Detection and mitigation of FDIAs is crucial to the safe and reliable operation of the system
- Targets sensors and aims to change measurement before used in state of charge (SoC) estimation



Healthy SoC curve



SoC following +100 mV FDIA



SoC following -100 mV FDIA

- Possible consequences of FDIA:



Power outages



Thermal runaway



Battery degradation



Increased costs



Injury



Equipment damage



Suboptimal control



Additional attacks



Unknown Probability of an Attack

- Hard to predict what vulnerabilities may be exploited
- New vulnerabilities can be discovered
- Different studies have differing results regarding the likelihood of cyberattacks

Zero-Trust Approach

- Do not assume a system or device is attack free
- Do not assume it is impossible to compromise a system
- Authentication, Authorization, and Validation can help with this

Defense-in-Depth Approach

- Add as many layers of protection to the system as reasonable / possible
- If one layer is compromised, backup layers exist stop threats
- Some layers may include: policy, physical, network, application, device

Fundamentals for battery energy storage system cybersecurity

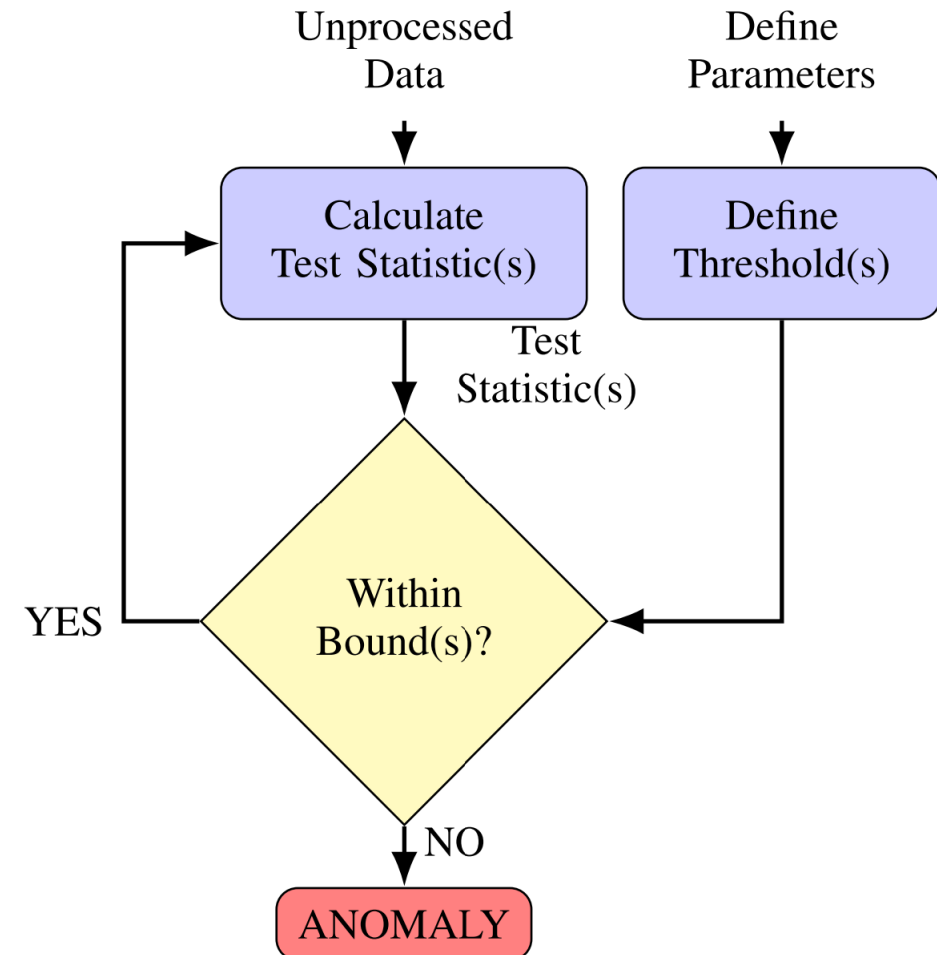
SOLUTION: MODEL-BASED ANOMALY DETECTION



Solution for: Security and Safety

Detects failures, faults, cyberattacks, anomalies

- **Step 1:** Accurate battery model to represent system
- **Step 2:** Estimator to estimate states and measurements
- **Step 3:** Compare estimated measurements to sensor readings, generate residual
- **Step 4:** Residual used by anomaly detector to flag anomalies



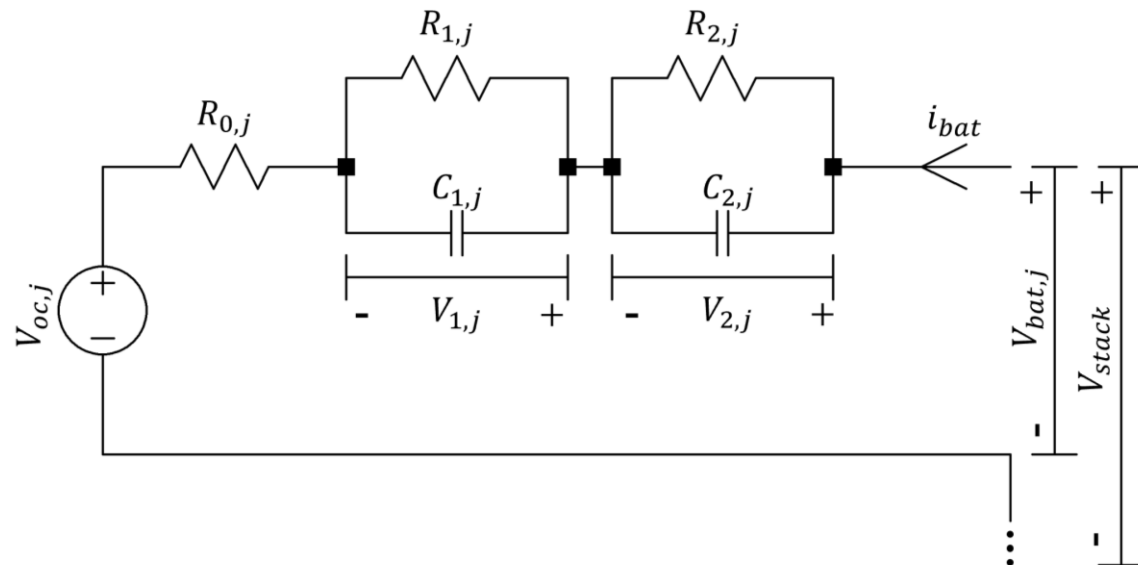
General model-based anomaly detector

STEP 1: SELECTED BATTERY MODELS



- Equivalent Circuit Model

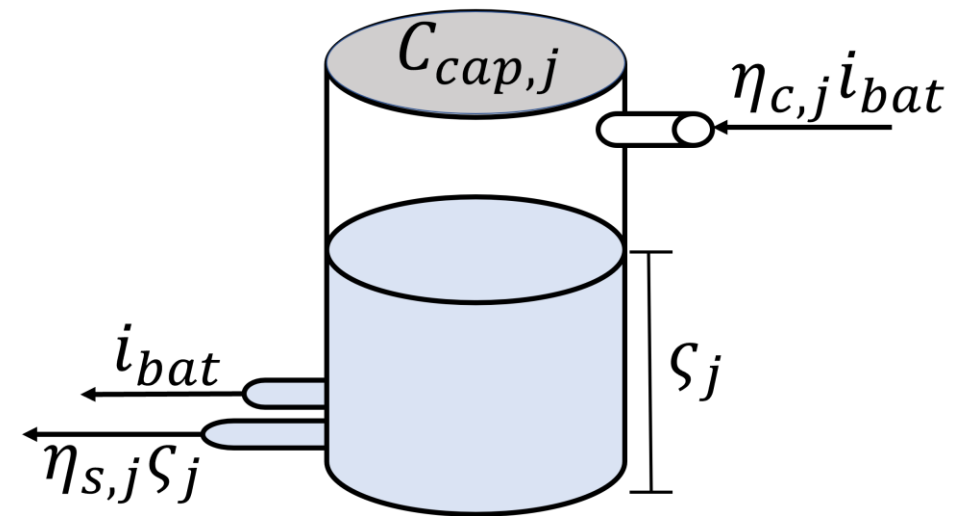
- Models the response of battery voltage (output) to the stack current (input)
- Good balance of accuracy and complexity
- Does not account for degradation



Equivalent circuit model

- Charge Reservoir Model

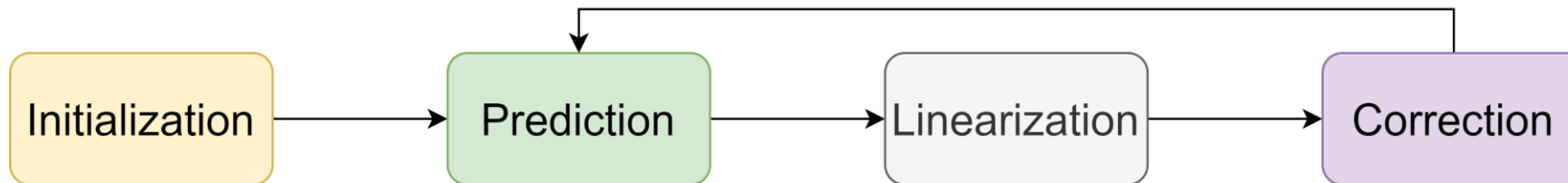
- Models charging and discharging as filling and draining of a cylindrical tank, respectively
- Required to supplement ECM, as it does not include SoC estimation



Charge reservoir model

STEP 2: SELECTED ESTIMATOR

- [Extended Kalman Filter](#)
- Used to estimate SoC and measurements
- Estimated measurements are compared to actual measurements to calculate the a priori measurement residual (used in detector)
- Compatible with nonlinear systems
- Theoretically less accurate than the unscented Kalman filter, but less computationally complex – we had similar results regardless of estimator

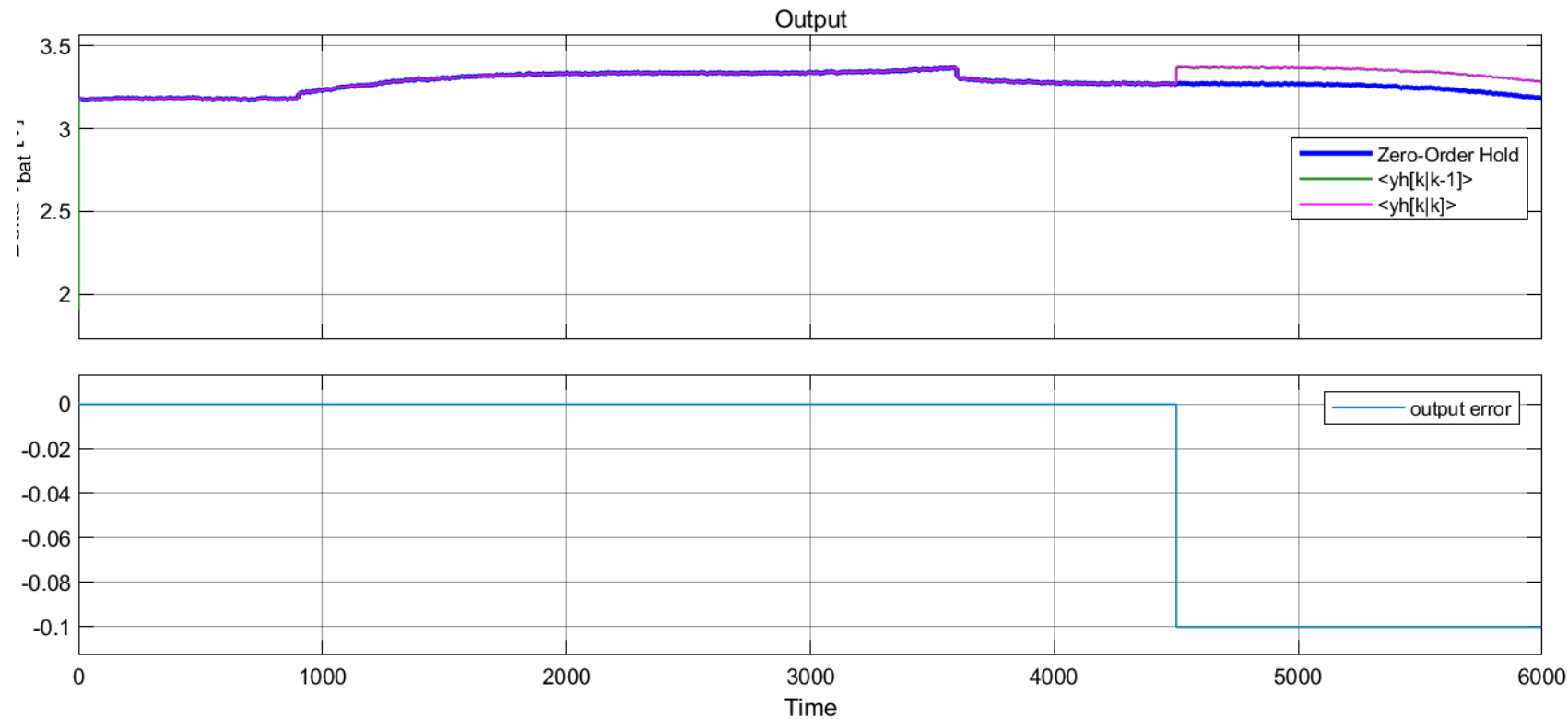


Simplified EKF Flowchart



STEP 3: SELECTED TEST STATISTIC

- Measurement residual
- Difference between the measurements captured by the sensors and the measurements estimated by the EKF



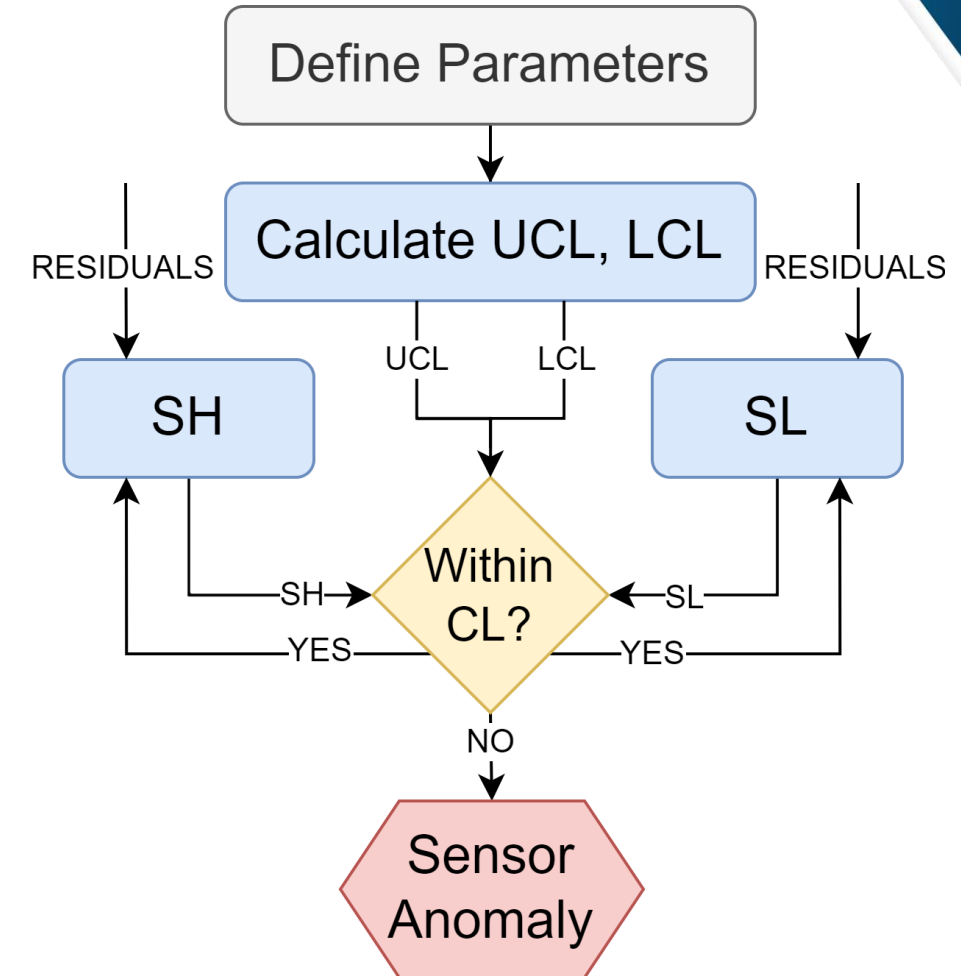
Battery voltage and residual



STEP 4: SELECTED DETECTOR

- [Cumulative Sum \(CUSUM\) Algorithm](#)
- Recursive sum applied for FDIA detection, using measurement residual
- In some cases, was able to identify the targeted sensor and classify the bias of the attack as positive or negative

General CUSUM Rules
Detection
If $SH > UCL$ or $SL < LCL \rightarrow$ FDIA Detected
Identification
If SH or SL of Sensor X diverges $\rightarrow$ FDIA injected in Sensor X
Classification
$SH > UCL \rightarrow$ Positively biased, $SL < LCL \rightarrow$ Negatively biased



Simplified CUSUM Flowchart

CASE STUDY: SIMULATED BATTERY STACK

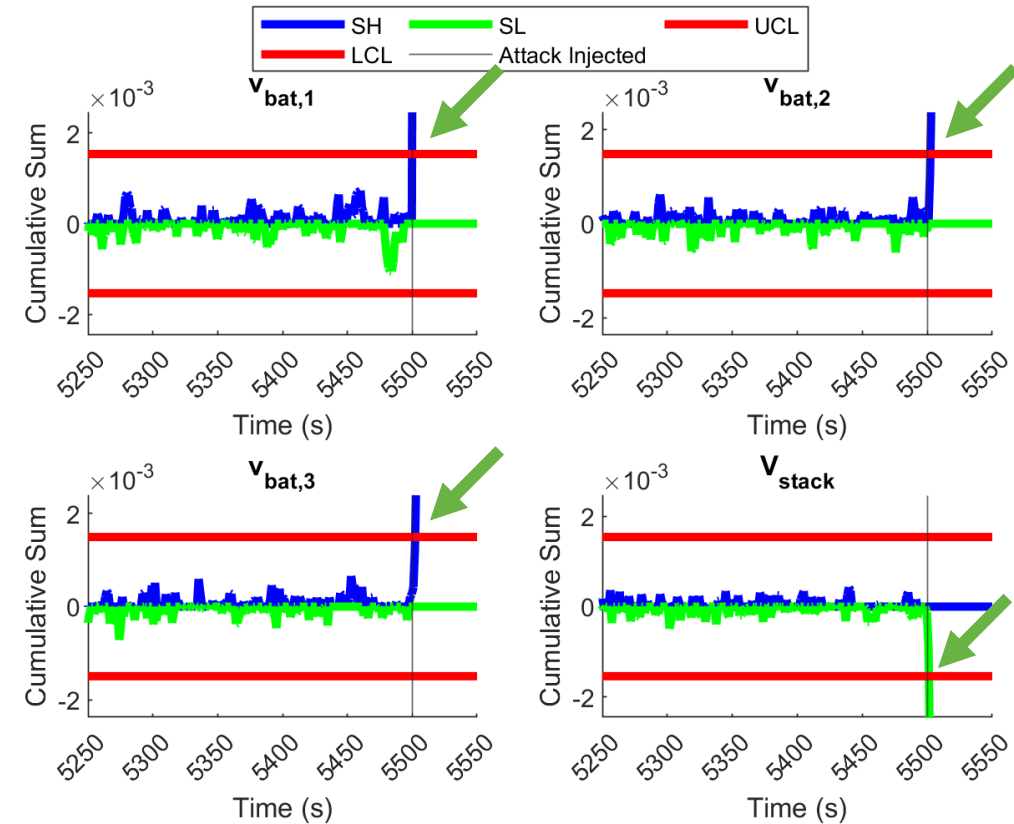


Simulation Setup

Battery Models	ECM + CRM
Estimator	EKF
Detector	CUSUM
Simulation time	8100 s
Simulations run	3200
Vulnerable Cells / Sensors	3 / 4
Attack range / resolution	$\pm 20 \text{ mV} / 153 \mu\text{V}$
Input Current	$\pm 4.0435 \text{ A}$

Batch Results

False Positive Rate	0%
Detection Rate	99.90%



Output CUSUM charts when a +20 mV attack was injected in the $v_{bat,1}$ sensor at 5500 s.

CONCLUDING THOUGHTS



A STRONG CYBERSECURITY POSTURE, IT TAKES ALL OF US!



- Collaborating with cybersecurity experts of different backgrounds
- Staying knowledgeable about known threats inside and outside of your field of study
- Workforce training and encouraging a culture of cybersecurity
- “See something, say something”
- Red, blue, and purple teaming
- **Proactively** implementing cybersecurity best practices





THANK YOU!

ACKNOWLEDGEMENTS

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PRACTICAL APPLICATIONS: MODEL-BASED ANOMALY DETECTION



- What can I use this method for?
- Anomaly detection (cybersecurity, thermal management, fault detection, failure detection)
- Statistical quality control, error detection, flagging shifting means
- Maybe be used with batteries or other systems with timeseries data, does not require a model

Requirements to use CUSUM

- 1) Timeseries of data
- 2) Known (or estimated) population mean
- 3) Known (or estimated) population standard deviation
- 4) Patience – tuning the parameters of the algorithm to suit your application can be time consuming

CUSUM Equations

High Sum (SH): $SH_i = \max(0, \bar{z}_i - \mu - \gamma\sigma_{\bar{z}} + SH_{i-1})$

Low Sum (SL): $SL_i = \min(0, \bar{z}_i - \mu + \gamma\sigma_{\bar{z}} + SL_{i-1})$

Upper Control Limit (UCL): $UCL = +h\sigma_{\bar{z}}$

Lower Control Limit (LCL): $LCL = -h\sigma_{\bar{z}}$

$\bar{z}_i$: sample mean of the test statistic

μ : population mean

γ : correction factor

$\sigma_{\bar{z}}$: population standard deviation

h : controls the width of the detection limits

REFERENCES AND PHOTO CREDIT



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